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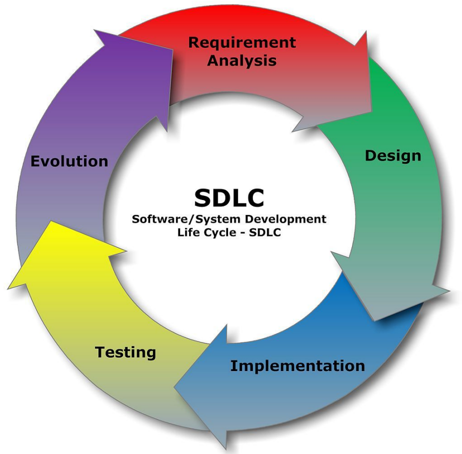
Introduction

SDLC and Agile

Software Development Life Cycle (SDLC)

- SDLC is the process and related activities of producing a software.
- There are different SDLCs; but all must include 5 fundamental activities:
  - Software specification (requirement analysis)
  - Software Design
  - Software implementation
  - Software validation (testing)
  - Software evolution (maintenance and further development)



- Some [popular SDLC models] are:
  - Waterfall, V-Model
  - Incremental, Iterative and Agile

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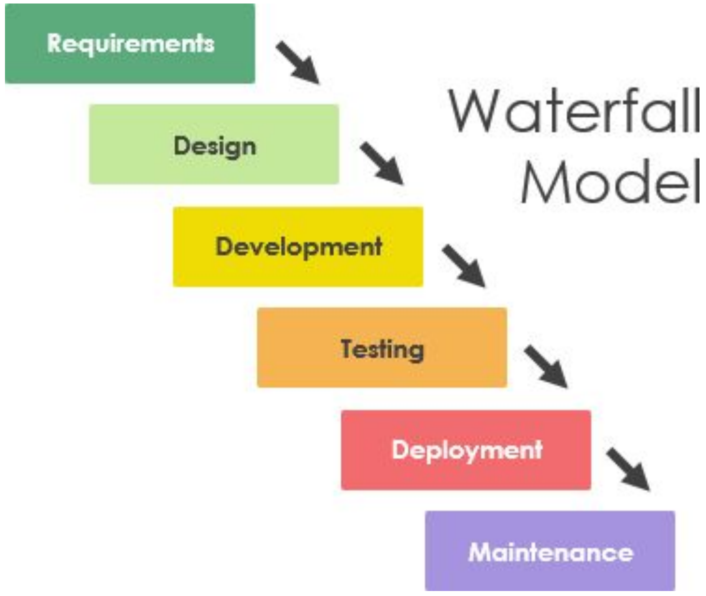
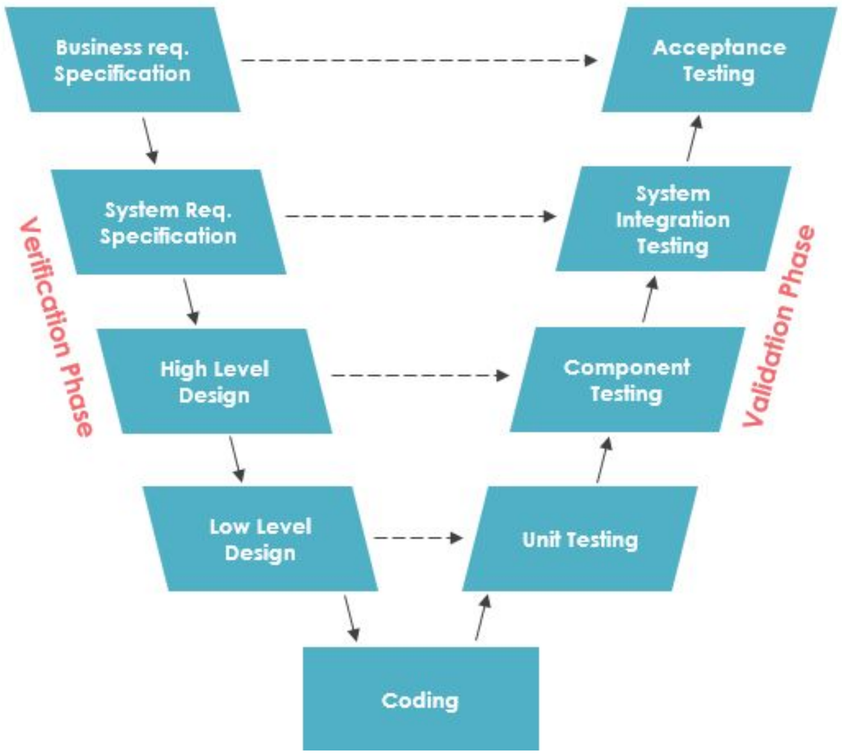
SDLC - Waterfall and V-Model

- In **Waterfall**, the whole process is divided into sequential phases and each phase needs to be completed before the next starts.
- **V-Model** is an extension of the waterfall model.
  - It is known as the **V**erification and **V**alidation Model



V-Model



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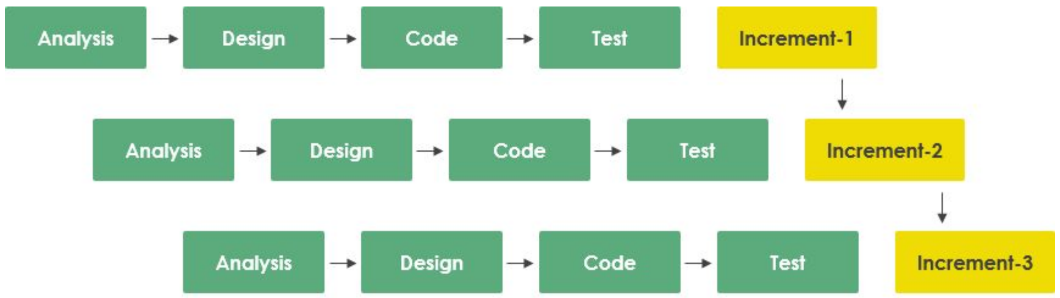
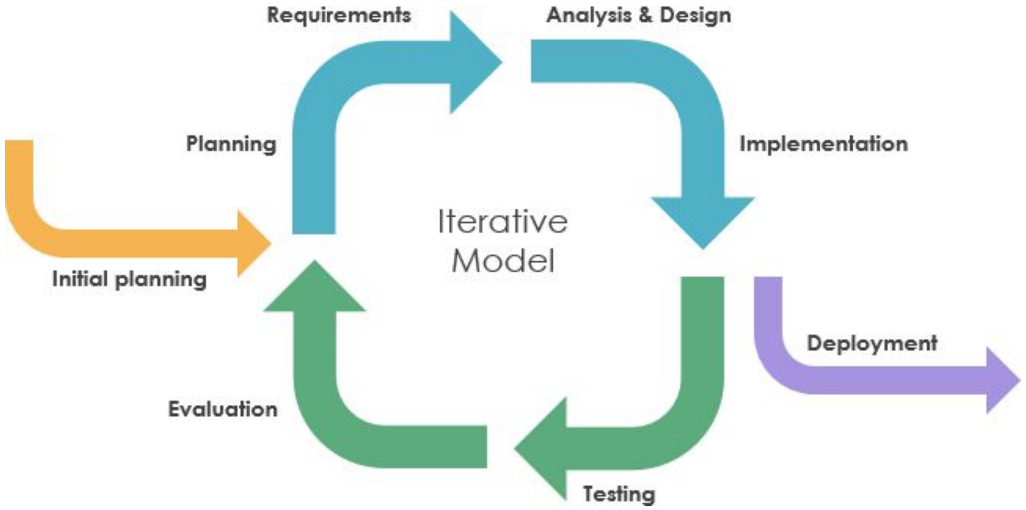
SDLC - Iterative and Incremental Models

- In the **iterative** model, development does not start with a full specification of requirements. it is suitable for scenarios where the final solution is hard to be defined.
- In the **incremental** model a software is designed, implemented and tested incrementally until the product is finished.This model combines waterfall model with the iterative philosophy.
- Both provide the development team with feedback from the previous increment or iteration.



Incremental Model



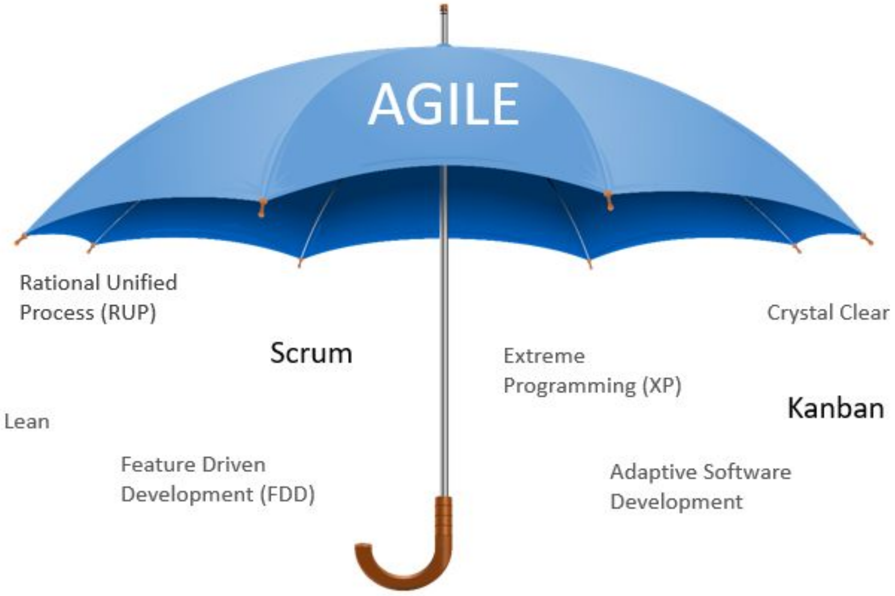
SDLC - Agile

❖ Is a collection of values and principles to constantly make improvements to the way we work. (Principles behind the Agile Manifesto)

The Agile Manifesto		
Individuals and interactions	over	Processes and tools
Working software	over	Comprehensive documentation
Customer collaboration	over	Contract negotiation
Responding to change	over	Following a plan

SDLC - Why Agile?

- Agile focuses on delivering high quality products and makes customers satisfied.
- Agile creates a work culture and mindset that enhances productivity
- Agile processes are more effective and efficient
- Agile principles ensure that team members are motivated
- The central theme of Agile is flexibility. Agile teams are responsive to change.



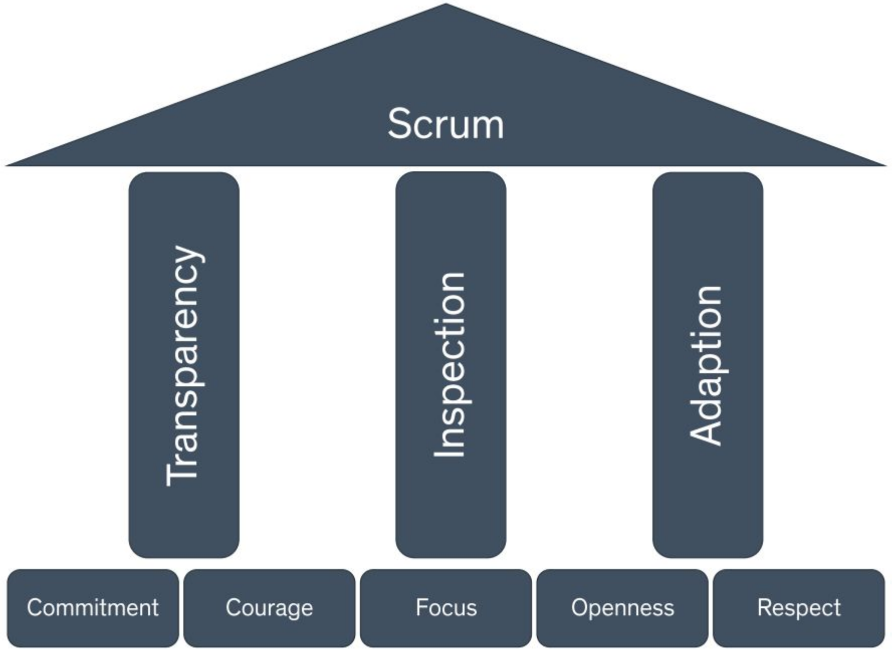
Agile - SCRUM

- Scrum is a process framework to manage work on complex products through adaptive solutions

- Scrum Values
  - Commitment
  - Focus
  - Openness
  - Respect
  - Courage

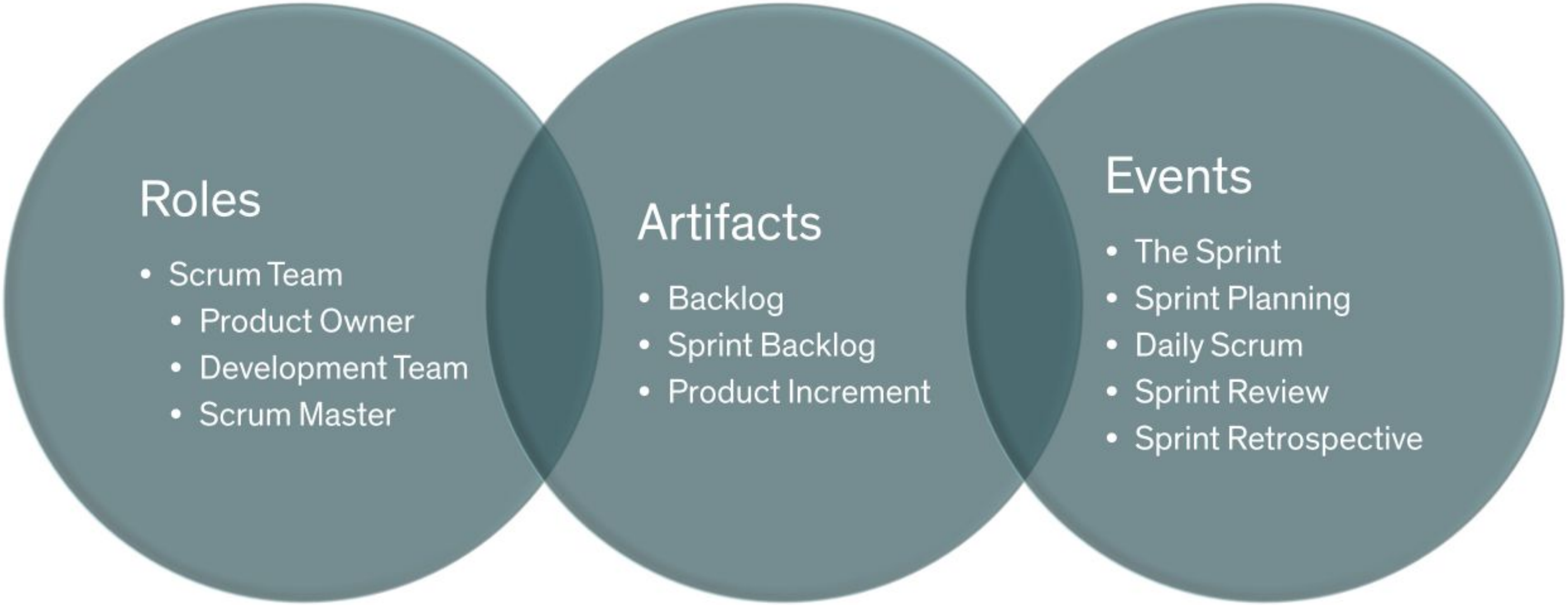
SCRUM - 3 Pillars

- Transparency
  - The process and work must be visible to those performing the work as well as those receiving the work.
- Inspection
  - The Scrum artifacts and the progress toward agreed goals must be inspected frequently to detect potentially undesirable problems.
- Adaptation
  - Continuous improvement and the ability to adapt based on the results of the inspection

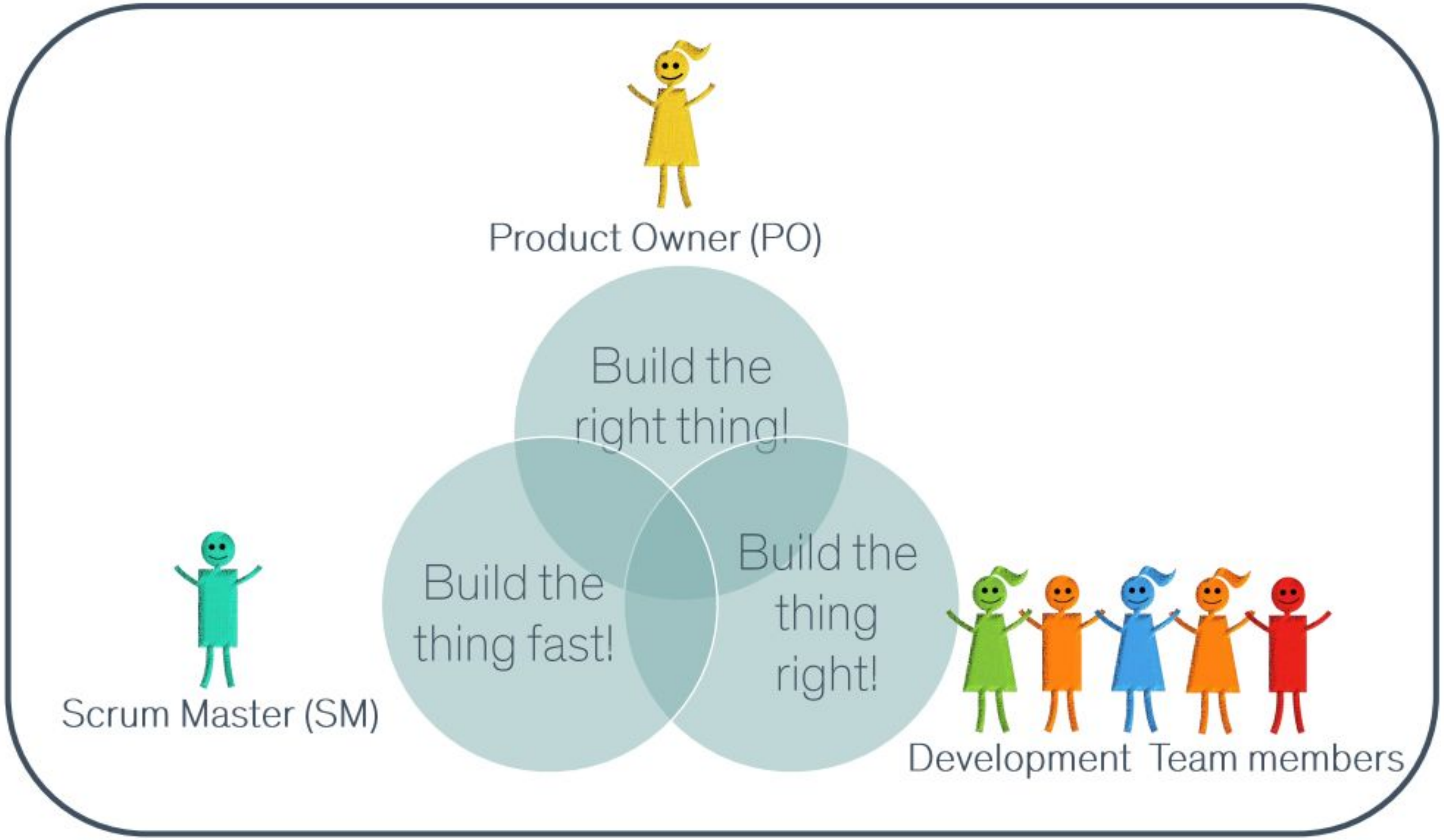


Scrum Framework

❖ Scrum Team, Artifacts and Events



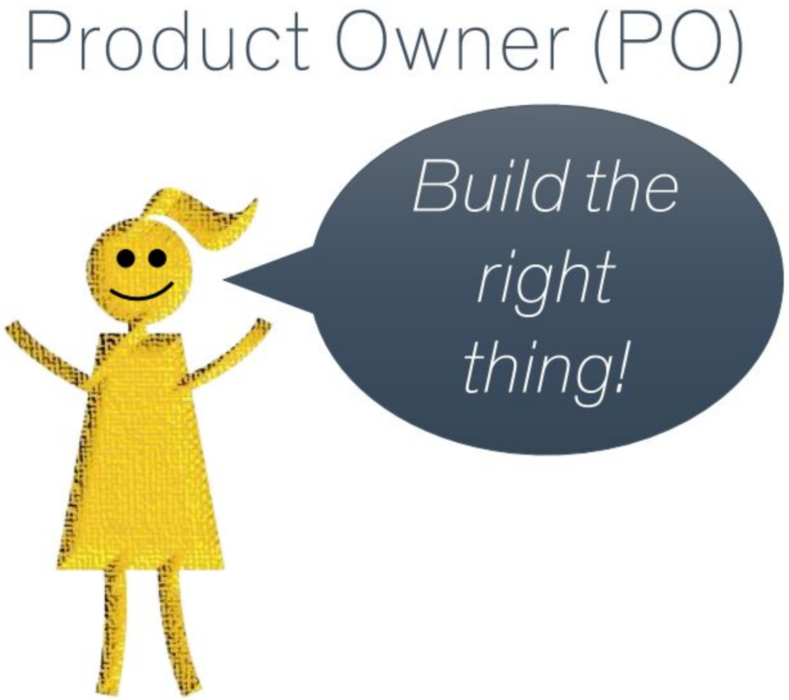
Scrum Team



Product Owner (PO)
Build the right thing!
Build the thing fast!
Build the thing right!
Scrum Master (SM)
Development Team members

Scrum Team - Product Owner

- Responsible to maximize value of the product
- Develops and communicates the Product Goal
- Defines the features of the product
- Prioritizes and maintains the product backlog
- Responsible for financial success of the product
- Ensures that the product backlog is transparent,
  visible and understood
- Manages and represents the needs of stakeholders
- Accepts or rejects works done by the development team



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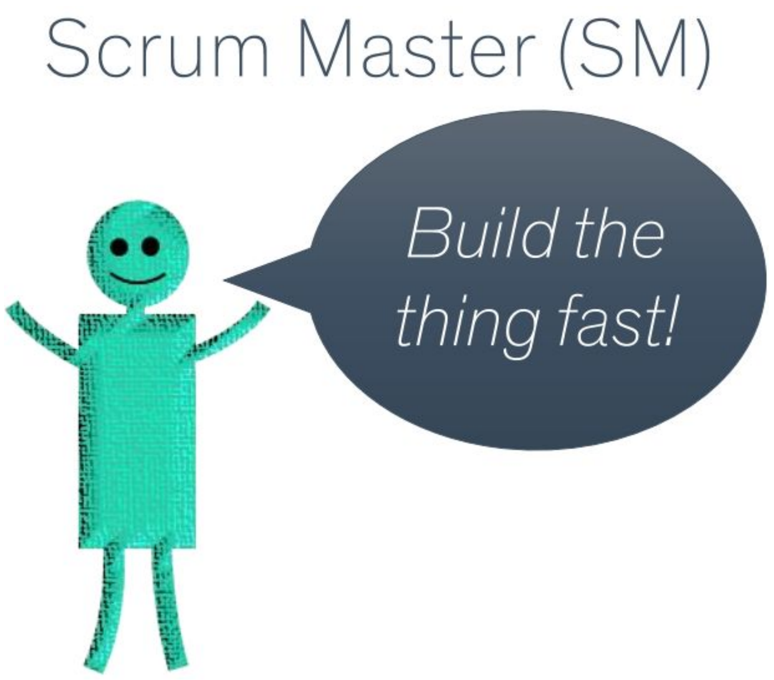
Scrum Team - Scrum Master

- ❖ Upholds Scrum values
- ❖ Enforces Scrum pillars
- ❖ Coaching the team members
- ❖ Helps everyone to understand the Scrum theory and practice
- ❖ Helps the Scrum team to focus on creating high-value Increments
- ❖ Removes impediments and blockers



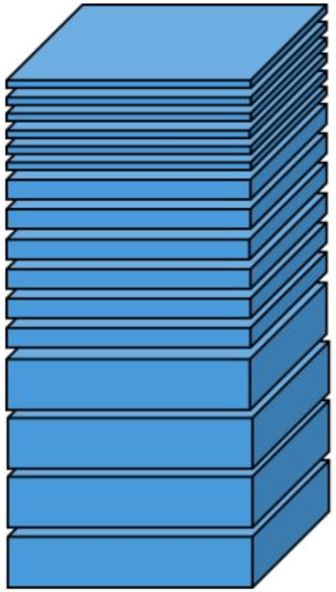
Scrum Team - Developers

- Self-organized
- Mutually accountable
- Have all the skills necessary to create value
- Typically 5 - 9 people
- Full-time employed
- Co-located; members must be close enough to each other (either physically or virtually)
  - They should be able to talk to each other within seconds or minutes.

Scrum Artifacts - Product Backlog

- ❖ User story: High-level definition of a requirement
- ❖ Task: Smallest unit of work which is estimable
- ❖ Is a prioritized list of all things that needs to be done within the project.
  - ➢ Definition of Done (DoD) is required.
- ❖ The single source of requirements for any changes to be made to the product
- ❖ A living document.
- ❖ Maintained by the product owner

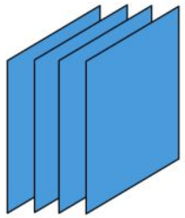
Product Backlog

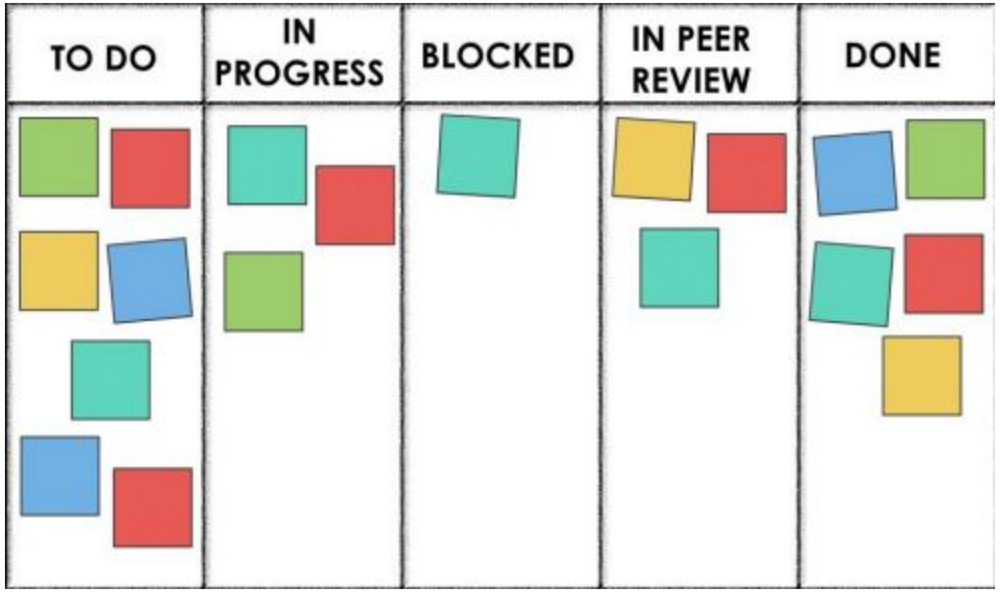


Scrum Artifacts - Sprint Backlog

- Is the set of Product Backlog items selected for the Sprint
- A forecast by the Development Team about what value will be in the next Increment and the work needed to deliver that value into a “Done” increment.
- The work that the Developers plan to accomplish during the Sprint in order to achieve the Sprint Goal

Sprint Backlog

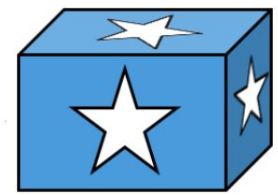




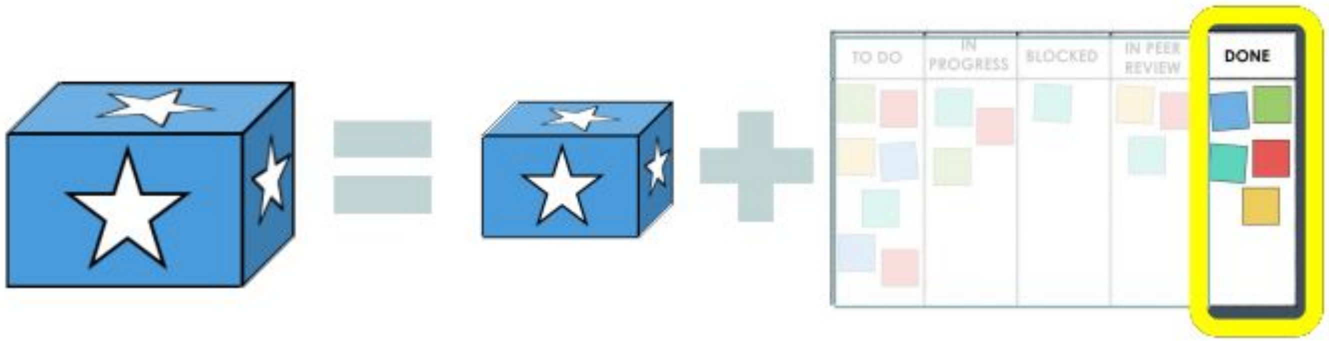
Scrum Artifacts - Product Increment

- Sum of all the Product Backlog items completed during a Sprint and the value of the increments of all previous Sprints

Product Increment

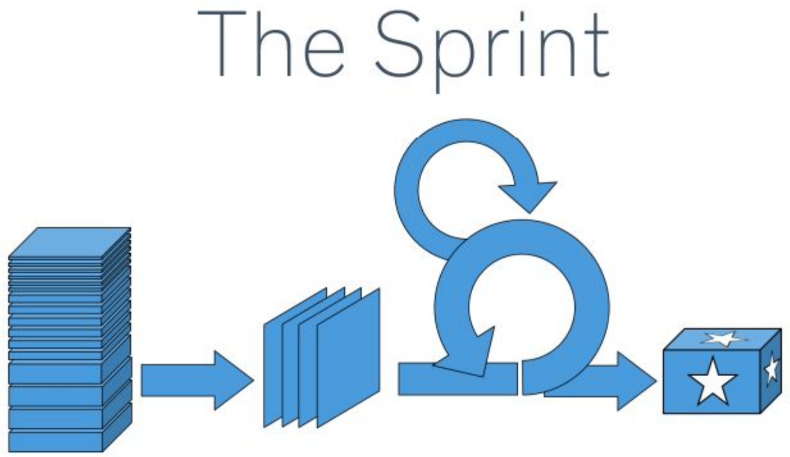


- An Increment is a concrete step toward the Product Goal
- Work cannot be considered as a part of an Increment unless it meets the Definition of Done.



Scrum Events - The Sprint

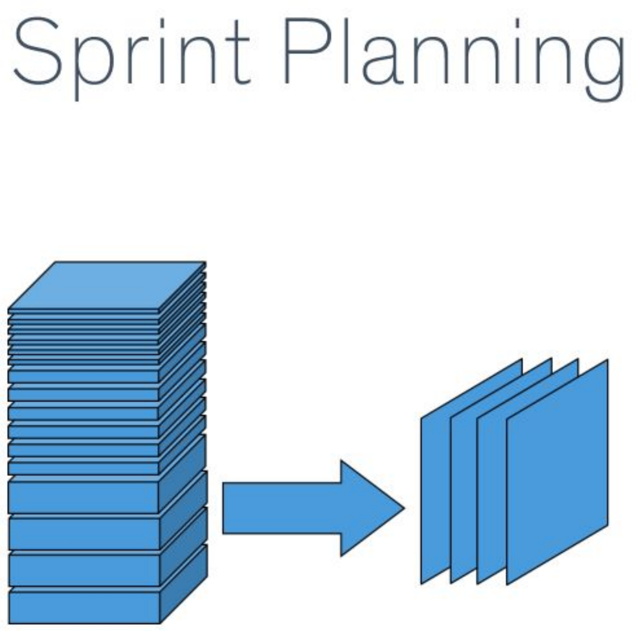
- A Timebox
  - Fixed length; 1 - 4 weeks
- Starts immediately after the conclusion of the previous Sprint.
- All the work necessary to achieve the Product Goal, including Sprint Planning, Daily Scrums, Sprint Review, and Sprint Retrospective, happen within Sprints.



- The purpose is creating a potentially releasable Product Increment
- Enables predictability by inspection and adaptation of progress toward a Product Goal

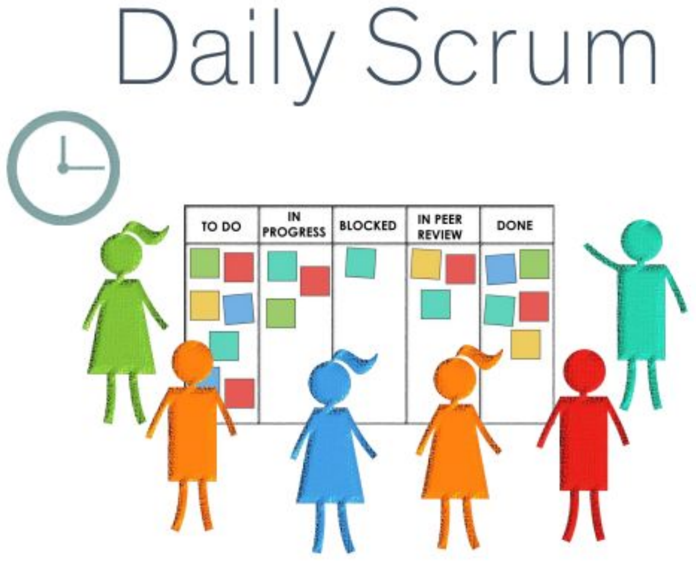
Scrum Events - Sprint Planning

- A Timebox
  - Max. 4 hours for a 2-week Sprint
- To plan the work to be performed during the Sprint
- To assess what the Development Team can accomplish during the Sprint
- To decide how the chosen work can be done and the Sprint goal can be achieved
- Is done by the Scrum Team. But the Scrum Team may also invite other people to attend Sprint Planning to provide advice.



Scrum Events - Daily Scrum

- A Timebox
  - 15 minutes
- To optimize work collaboration and performance
- To inspect progress towards completing the Sprint Goal and Sprint Backlog
- To improve communications, identify impediments and promote quick decision-making and help each other



- What did I do yesterday that helped the Development Team meet the Sprint Goal?
- What will I do today to help the Development Team meet the Sprint Goal?
- Do I see any impediment that prevents me or the Development Team from meeting the Sprint Goal?

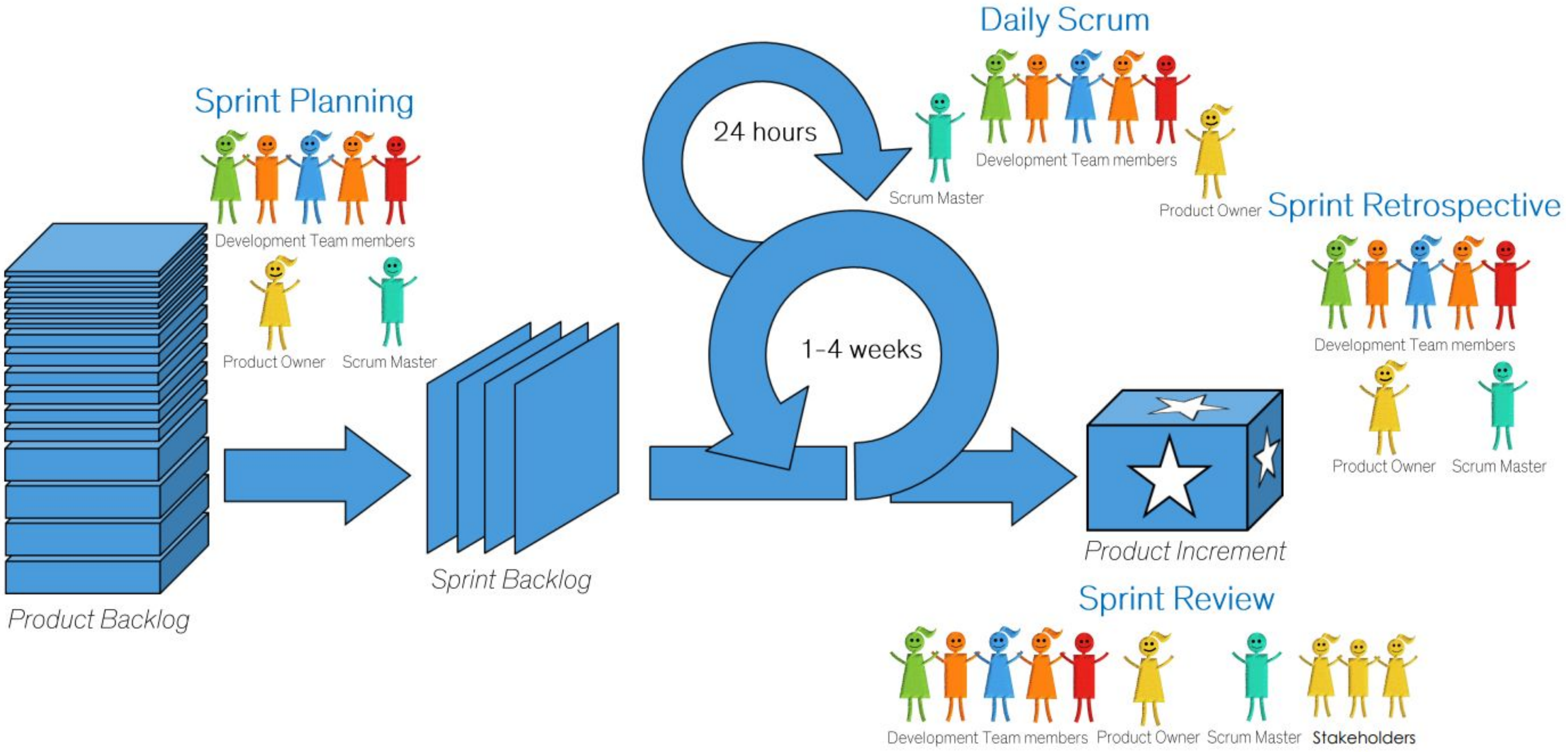
Scrum Events - Sprint Review

- A Timebox
  - Max. 4 hours for a 4-week Sprint
- To inspect the Product Increment
- To adapt the Backlog
- To collaborate what was done during the Sprint between the Scrum Team and stakeholders.
  - The Scrum Team presents results of its work to stakeholders and progress toward the Product Goal is discussed.

Scrum Events - Sprint Retrospective

- A Timebox
  - Max. 3 hours for a 4-week Sprint
- To inspect how the Sprint went with regards to the team, collaboration, processes and tools
- To create a plan for improvement on how the team conducts its work
- To conclude the Sprint

Scrum in a Nutshell



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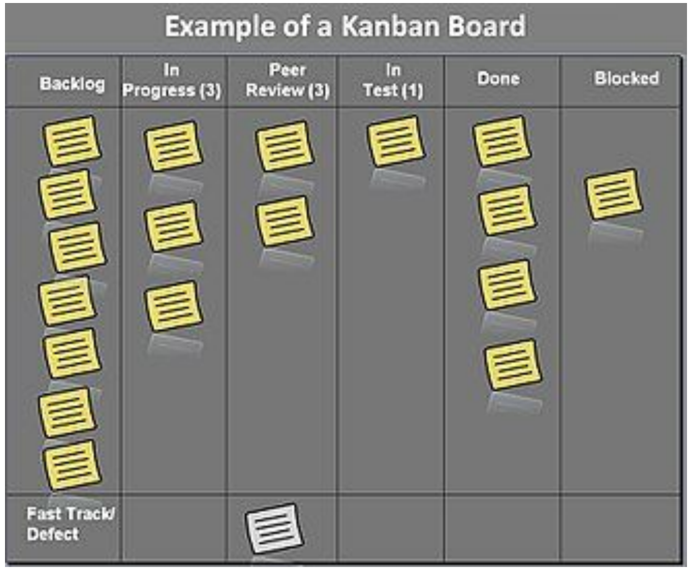
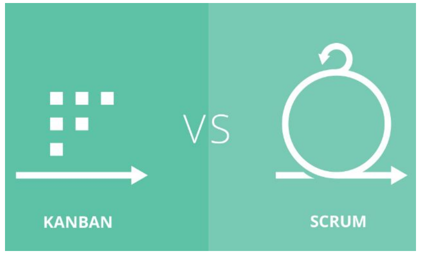
Agile - Kanban

- An agile framework
- Visualized the workflow using a **kanban** board
- Emphasizes on limiting the amount of work in progress.
- Time-boxed iterations are optional
- No particular item size is required, estimation is optional
- New items can be added whenever capacity is available
- The Kanban board may be shared by multiple teams or individuals
- Doesn't have definite roles
- Prioritization of tasks is optional
- **Scrumban**: Best practices from Scrum and Kanban



🔗 What Is Kanban?



🔗 Scrum vs Kanban

Introduction to SDLC and Agile

- Some useful links
  - Software Development Life Cycle Models
  - What is a Software Process Model?
  - Agile vs Waterfall
  - Manifesto for Agile Software Development
  - Principles behind the Agile Manifesto
  - What is Agile?
  - What is Agile?
  - Scrum Guide
  - What Is Scrum?
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